

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Year: 2018

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 26 April 2019

Time: 8.00-10.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 10 pages (including this page) and they are worth 85 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.
- In part 1 and 2, enter your answer in the answer sheet on the LAST PAGE.
- In part 3, show all the steps of your solution with answers in the provided space underneath of each problem.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

### Part 1 (20 points) True/False

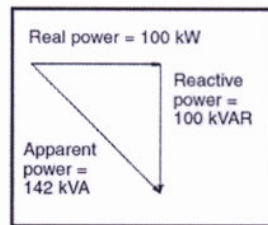
Instruction: Following questions, mark "X" for the correct answer, "True" or "False" in the provided answer sheet (see LAST PAGE).

1. The Norton equivalent circuit cannot be converted to Thevenin equivalent circuit.
2. The voltage of independent source always provides constant voltage which is not affected by any load current.
3. The voltage of dependent source is controlled by either a current or a voltage in circuit.
4. The power factor is  $\omega$ .
5. The positive reactance is a capacitive reactance.
6. Inductive reactance is decreased by increasing frequency.
7. Kirchhoff's Current Law is applied for the nodal analysis at each node in circuit.
8. The unit of real power is in W.
9. The more quality factor, the less frequency bandwidth.
10. Resonance occurs when the reactive capacitance equals to reactive resistance.
11. The superposition theorem states that the current and voltage for any element in the circuit is the sum of the currents and voltages produced by each source independently.
12. Equivalent Impedance of Norton and Thevenin equivalent circuit are the same.
13. The primary and secondary currents of a transformer are related by the ratios of turns.
14. Ohm's law is applied in the voltage and current sources conversion.
15. In Mesh analysis, the voltage polarity of each R, L and C elements depends on the direction of distinct current of each loop.
16. In Node analysis, 2 nodes provide 2 linear equations.
17. In a Y-connected load, the 220 V is available from line-to-neutral and the 380 V is available from line-to-line.
18. The current and voltage is in phase, it means that the power factor of circuit is unity.
19. Complex conjugate refers to the opposite sign of imaginary part.
20. High quality factor in series RLC circuit provides narrow bandwidth.

## Part 2 (40 points) Multiple choice

Instruction: Mark the correct answer for the following questions in the provided answer sheet (see LAST PAGE).

1. What is the power factor of power triangle below?

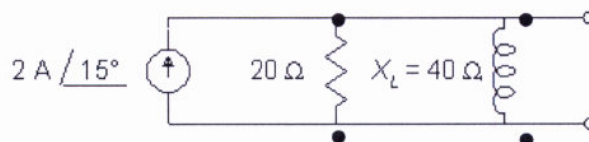


- (a) 0.6 (b) 0.7 (c) 0.8 (d) 1.0

2. Convert the current source to a voltage source. Given  $I_s = 2\sin(\omega t + 30^\circ)$  A and parallel impedance  $Z = 100 + j0$ . What is the new voltage source?

- (a)  $V_s = 20 \angle 60^\circ$  (b)  $V_s = 200 \angle 60^\circ$  (c)  $V_s = 0.2 \angle 30^\circ$  (d)  $V_s = 200 \angle 30^\circ$

3. Convert the current source to a voltage source. What is the angle of the current?



- (a)  $41.6^\circ$  (b)  $25.5^\circ$  (c)  $27.1^\circ$  (d)  $24.3^\circ$

4. The superposition theorem in AC network analysis, which one of the following statements is true?

- (a) Voltage sources are replaced by open circuits.  
 (b) Voltage sources are replaced by short circuits.  
 (c) Current sources are replaced by short circuits.  
 (d) All impedances are replaced by the complex conjugates.

5. If  $Z_{th} = 100 + j20 \Omega$ , what is the load impedance for maximum power transfer.

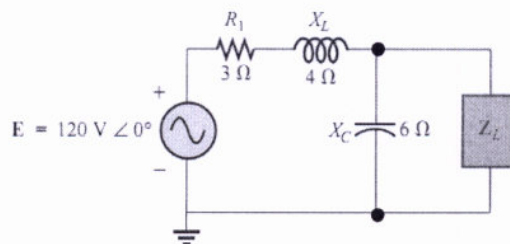
- (a)  $50 - j10 \Omega$  (b)  $100 - j20 \Omega$   
 (c)  $-100 - j20 \Omega$  (d)  $10 + j20 \Omega$

6. If  $S = 5 - j10 \text{ VA}$ , which of the following is true?

- (a) Power factor is greater than one.  
 (b) Inductive reactance must be greater than capacitive reactance.  
 (c) It is pure resistive circuit  
 (d) The average power is equal to 5 W

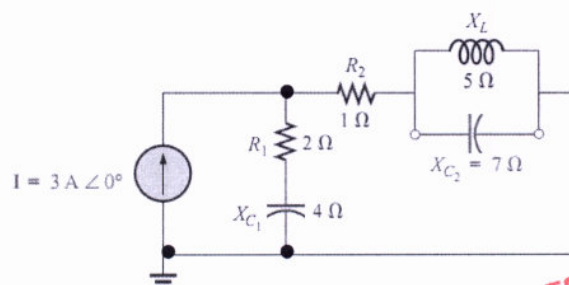


7. From figure below, what is the Thevenin equivalent impedance  $Z_{th}$  at terminal?



- (a)  $(R_1 // jX_L) - jX_C$  (b)  $(R_1 + jX_L) // (-jX_C)$   
 (c)  $(R_1) // (jX_L) // (-jX_C)$  (d)  $R_1 + jX_L - jX_C$

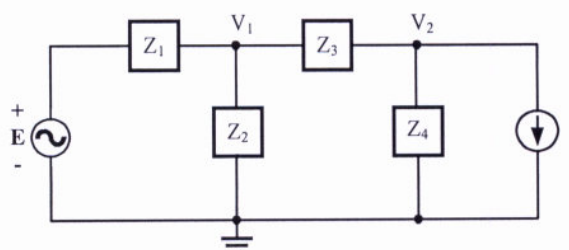
8. From figure below, what is the Thevenin equivalent impedance  $Z_{th}$  external to  $X_{C2}$  (after removing  $X_{C2}$  out)?



- (a)  $(R_1 - jX_C + R_2) // (jX_L)$  (b)  $(R_1 - jX_C) // (R_2 + jX_L)$   
 (c)  $(R_1 - jX_C + R_2 + jX_L)$  (d)  $(R_1 - jX_C) // (R_2) // (jX_L)$

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9. From the circuit below, what is KCL equation at node  $V_2$ ?

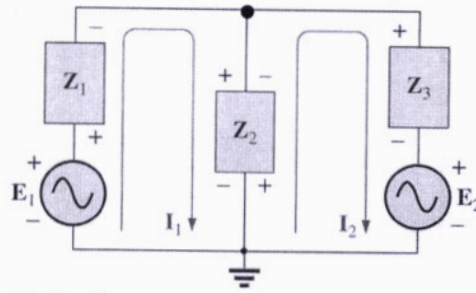


- (a)  $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$   
 (b)  $(V_2 + V_1)/Z_3 + V_2/Z_4 - I = 0$   
 (c)  $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$   
 (d)  $(V_2 - V_1)/Z_3 + V_2/Z_4 + I = 0$

10 From the previous problem, what is KCL equation at node  $V_1$ ?

- (a)  $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$   
 (b)  $(V_1 - E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$   
 (c)  $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$   
 (d)  $(V_1 - E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$

11. From the circuit below, what is KVL equation at mesh  $I_1$ ?

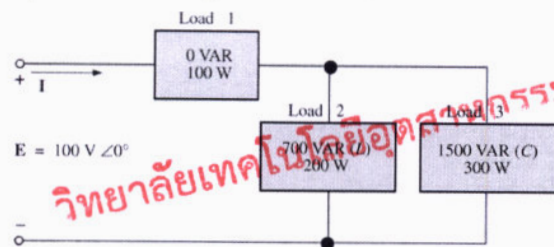


- (a)  $E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$  (b)  $E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$   
 (c)  $-E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$  (d)  $-E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$

12. From previous problem, what is KVL equation at mesh  $I_2$ ?

- (a)  $E_2 + Z_3 I_2 + Z_2 (I_1 - I_2) = 0$  (b)  $E_2 + Z_3 I_2 + Z_2 (I_2 - I_1) = 0$   
 (c)  $-E_2 + Z_3 I_2 + Z_2 (I_1 - I_2) = 0$  (d)  $-E_2 + Z_3 I_2 + Z_2 (I_2 - I_1) = 0$

13. What is the total real power of the system below?



- (a) 600 VA (b) 600 W  
 (c)  $600 + j2200$  VA (d)  $600 - j2200$  VAR

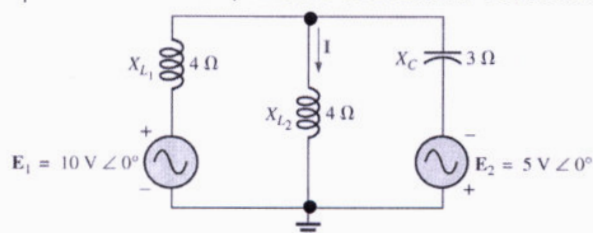
14. What is the reactive power of the previous problem.

- (a) 800 VAR (b) 800 W  
 (c)  $600 + j2200$  VA (d)  $600 - j2200$  VAR

15. What is the apparent power of the previous problem.

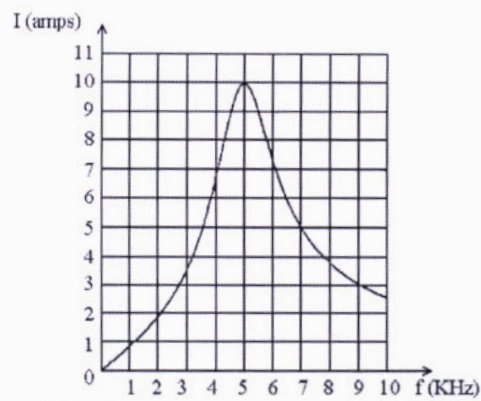
- (a)  $600 + j800$  VA (b)  $600 - j800$  VA  
 (c)  $600 + j2200$  VA (d)  $600 - j2200$  VA

16. Following the superposition theorem, what is the current  $I$  of source  $E_2$ ?

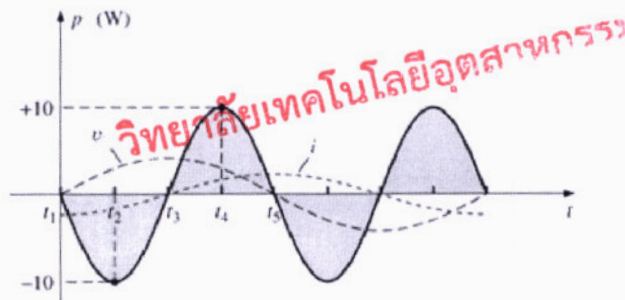


- (a)  $2.5 \angle 90^\circ$  (b)  $2.5 \angle 187.4^\circ$  (c)  $2.5 \angle 179.7^\circ$  (d)  $2.5 \angle 217.9^\circ$

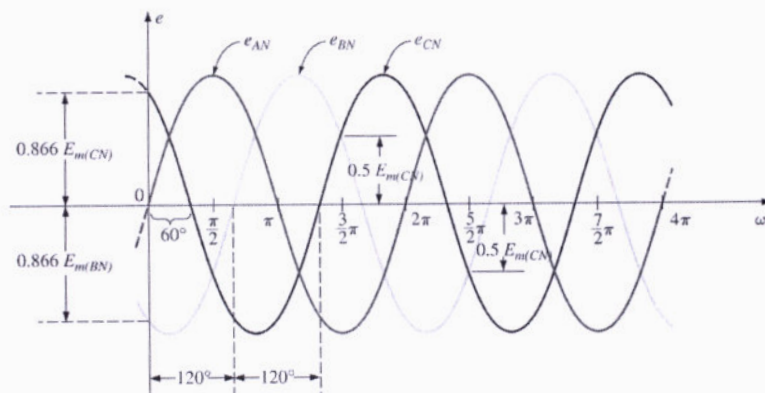
17. The curve below describes a series resonant circuit. What is an estimated bandwidth of circuit?



- (a) 1 kHz      (b) 2 kHz      (c) 3 kHz      (d) 4 kHz
18. What is the load type of following curve?

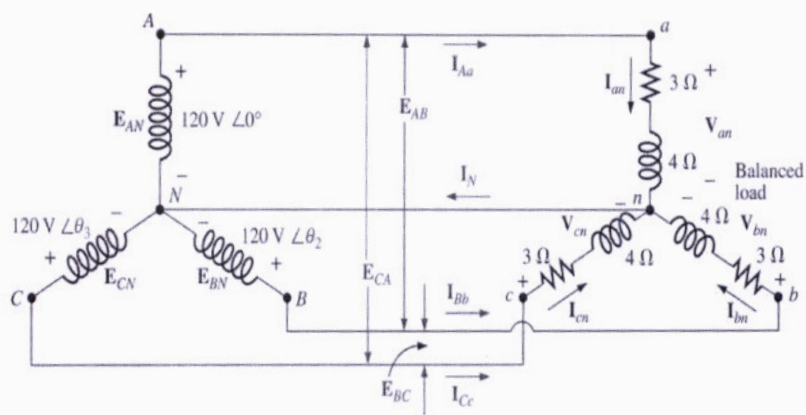


- (a) Capacitive Load      (b) Inductive Load  
(c) Resistive Load      (d) Hysteresis Load
19. From the time domain signal of three-phase generator in picture below, what is the  $e_{AN}$  equation?



- (a)  $e_{AN} = E_{m(AN)} \sin(\omega t + 120)$       (b)  $e_{AN} = E_{m(AN)} \sin(\omega t + 0)$   
(c)  $e_{AN} = E_{m(AN)} \sin(\omega t - 240)$       (d)  $e_{AN} = E_{m(AN)} \sin(\omega t - 120)$

20. Find the magnitude of the line voltages  $E_L$  and magnitude of line currents  $I_L$  of figure below.



(a)  $E_L = 208\text{V}$  ,  $I_L = 24\text{A}$

(b)  $E_L = 120\text{V}$  ,  $I_L = 24\text{A}$

(c)  $E_L = 108\text{V}$  ,  $I_L = 14\text{A}$

(d)  $E_L = 208\text{V}$  ,  $I_L = 14\text{A}$

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## Part 3 (25 points)

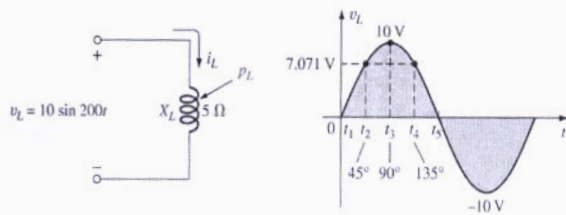
Instruction: Show the mathematical expression and answers of following problems.

1. From figure below,

a. Find the instantaneous power level for the inductor at time  $t_1$  to  $t_5$ .

b. Find the average power of inductor.

c. Find the energy stored in the inductor.

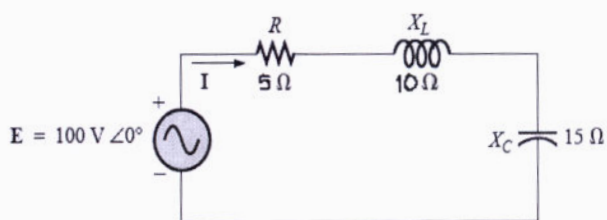


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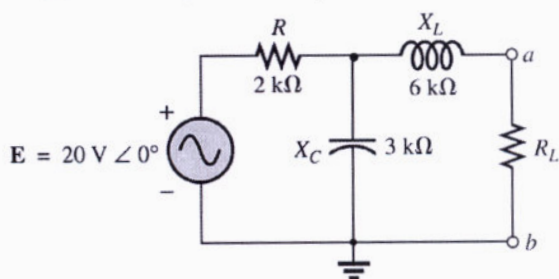
2. From the network in Figure below;

(a) Find the total number of watts (P), volt-amperes reactive (Q), and volt-amperes (S),

(b) Sketch the power triangle.

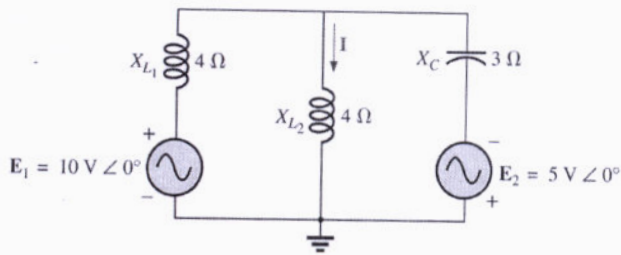


3. From figure below, find the voltage across  $R_L = 10 \Omega$  by using Thevenin theorem.

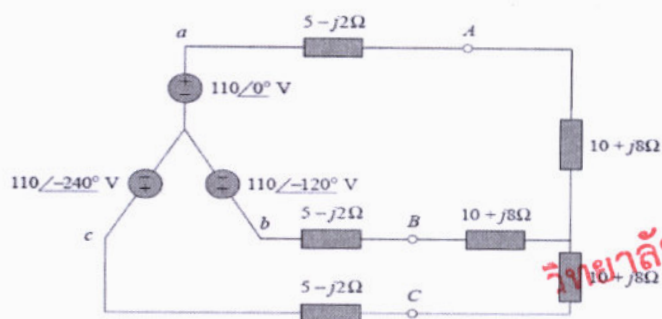




4. Find voltage across  $L_2$  and current  $I$  in circuit in figure below by using Mesh analysis



5. The star load configuration is connected with the three-phase generator as shown in picture below, Find the current in each branch of the circuit.



- Determine the current of branch A
- Determine the current of branch B
- Determine the current of branch C

Instructors: Assoc. Prof. Dr. Titipong Lertwiriaprapa  
and Asst. Prof. Dr. Kittisak Phaebua

Answer Sheet

Name: \_\_\_\_\_ ID: \_\_\_\_\_

Subject: 341152 Electric Circuits II

Part 1

|     | True | False |
|-----|------|-------|
| 1.  |      |       |
| 2.  |      |       |
| 3.  |      |       |
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| 18. |      |       |
| 19. |      |       |
| 20. |      |       |
|     |      |       |

Part 2

|     | a | b | c | d |
|-----|---|---|---|---|
| 1.  |   |   |   |   |
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| 20. |   |   |   |   |

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